

Name: _____ Student ID number: _____

MIDTERM

No books or notes are allowed. **100 Points, 120 Minutes.**

1.(5 points) Prove that $f(z) = \frac{1}{z}$ does not have a primitive in the punctured disc $\mathbb{D}^* = \{z \in \mathbb{C} : 0 < |z| < 1\}$.

2.(5 points) Find the number of solutions of $z^{2020} - 2019z^{2018} + z^{2017} + z^{2016} + 2015 = 0$ in the unit disc $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$.

3.(10 points) Let b be a complex number. Evaluate $\int_0^{2\pi} \log \left| \frac{b}{2020} - e^{i\theta} \right| d\theta$.

4.(15 points) Let $x > 0$. Evaluate

$$\frac{1}{2\pi i} \int_{\operatorname{Re}(z)=2020} \frac{x^z}{z} dz,$$

where the vertical line $\operatorname{Re}(z) = 2020$ takes the direction from $2020 - i\infty$ to $2020 + i\infty$.

5.(20 points) Let $k \in \mathbb{N}$. Evaluate $\zeta(2k) = \sum_{n=1}^{\infty} \frac{1}{n^{2k}}$ as follows.

(a) Show that $\zeta(2k) = -\frac{1}{2} \operatorname{Res}_{z=0} \frac{H(z)}{z^{2k}}$, where $H(z) = \frac{2\pi i}{e^{2\pi i z} - 1}$. (Hint: first apply the residue formula on the positively oriented boundary of the square with vertices at the points $(N+1/2)(\pm 1 \pm i)$).

(b) Show that $\frac{z}{e^z - 1}$ has a power series expansion $\sum_{l=0}^{\infty} \frac{B_l}{l!} z^l$ for specific values of z . Relate the value of $\zeta(2k)$ with the coefficients B_l . In particular, find the explicit value of $\zeta(2)$.

6.(10 points) (a) Assume that $f(z)$ is a non-constant holomorphic function in $\{1 < |z| < \infty\}$, and the limit $\lim_{z \rightarrow \infty} f(z) = f(\infty)$ exists and is finite. Prove that $f(\infty) = \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\theta}) d\theta$, where $R > 1$, and that $|f(z)|$ cannot obtain its maximum in $\{|z| > 1\}$.

(b) Apply (a) to prove that any holomorphic function on the Riemann sphere $\bar{\mathbb{C}}$ must be constant.

7.(10 points) Let Ω be an open connected subset of $\mathbb{C}$ and $\{f_n\}$ a sequence of injective holomorphic functions on Ω . Assume that $\{f_n\}$ converges uniformly on every compact subset of Ω to a holomorphic function f , prove that f is either injective or constant.

8.(15 points) Consider the polynomial $P(z) = z^{2n} + 2020z^{2n-1} + 2020$. Find the numbers of solutions of $P(z) = 0$, whose real parts are positive, zero and negative respectively.

9.(10 points) Let $f(z)$ be a holomorphic function on an open set containing the closed disc $D = \{|z| \leq R\}$. Denote $M(r) = \max_{|z|=r} |f(z)|$, $A(r) = \max_{|z|=r} \operatorname{Re} f(z)$, where $0 \leq r \leq R$. Prove that for any $0 \leq r < R$, we have

$$M(r) \leq \frac{2r}{R-r} A(R) + \frac{R+r}{R-r} |f(0)|.$$